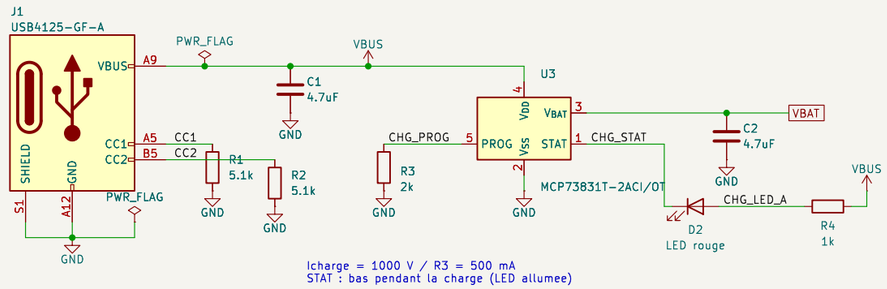
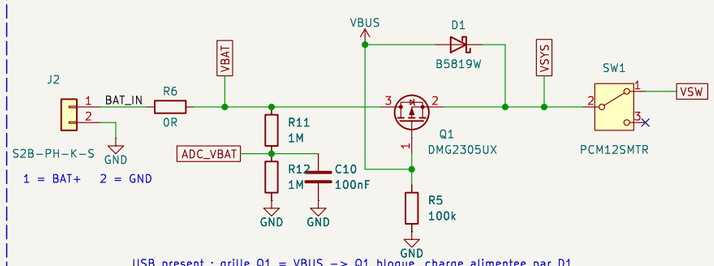
1 – Entree USB-C et charge Li-ion (MCP73831, 500 mA)



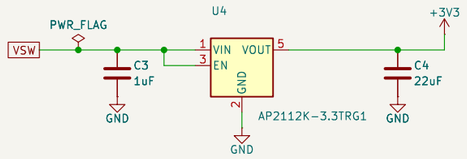
Icharge = 1000 V / R3 = 500 mA
STAT : bas pendant la charge (LED allumee)

2 – Batterie, partage de charge et interrupteur

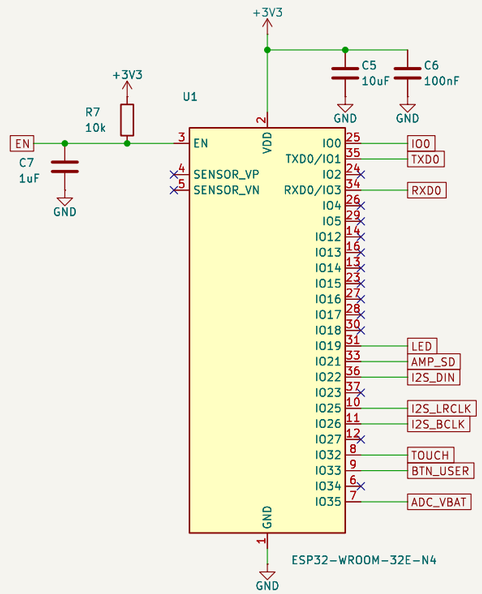


USB present : grille Q1 = VBUS -> Q1 bloque, charge alimentee par D1
USB absent : grille a 0 V par R5 -> Q1 passant, charge sur batterie

3 – Regulation 3,3 V (AP2112K, 600 mA)

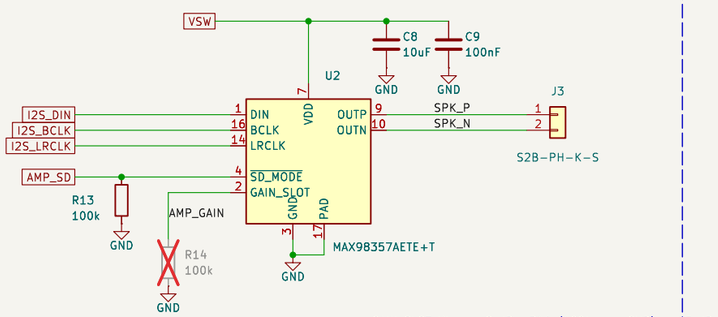


4 – ESP32-WROOM-32E (Bluetooth Classic A2DP + BLE)



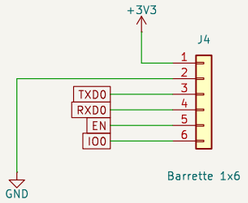
Broches de demarrage IO2, IO5, IO12, IO15 :
laissees libres. IO0 : bouton BOOT + J4 seulement.

5 – Ampli classe D I2S MAX98357A + haut-parleur



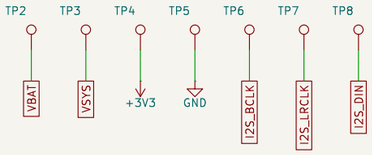
GAIN_SLOT libre = 9 dB ; R14 (100k vers GND) = 15 dB
SD_MODE : 0 V = arret (R13) ; 3,3 V = canal gauche

7 – Programmation UART (J4)

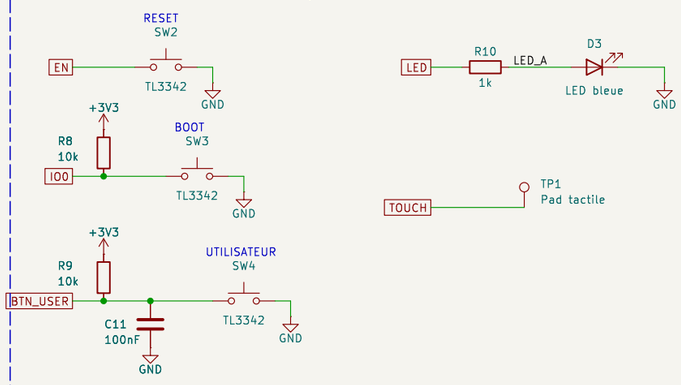


1 +3V3
2 GND
3 TXD0 (sortie ESP32)
4 RXD0 (entree ESP32)
5 EN
6 IO0

8 – Points de test



6 – Interface : boutons, LED, pad tactile



Oreillette Bluetooth v1 – carte de validation fonctionnelle
Chaine audio : A2DP -> ESP32 -> I2S -> MAX98357A -> HP ; retour BLE vers l'application.
Voir VERIFICATION.md pour les brochages controles et les ecarts avec la netlist d'origine.

Schema genere par tools/gen_sch.py puis verifie par ERC (kicad-cli)
ESP32-WROOM-32E + MAX98357A + chargeur Li-ion

Sheet: /
File: oreillette_v1.kicad_sch

Title: Oreillette Bluetooth v1 – carte de validation

Size: A3	Date: 2026-11-11	**Rev: 1.0**
KiCad E.D.A. 9.0.8		Id: 1/1